CNNP- Skill 2 (2520090047 - M. Harshika)

THREE-POINT CAPTURE – Analyze packet captures from three points in a network path to understand packet flow and identify performance differences.

STAGE 1 — Install tools

Open Ubuntu and type the following commands in the Ubuntu terminal

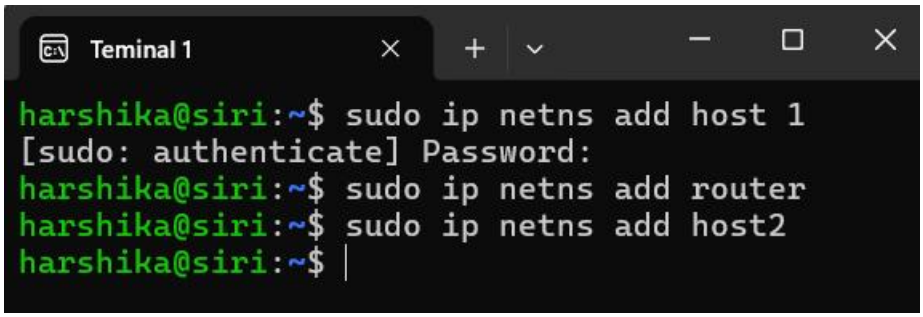
sudo apt update
sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd

```
harshika@siri:~$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:2 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [717 kB]
Hit:3 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [157 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [220 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [868 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [72.5 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [62.7 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [586 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [122 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [20.0 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [3728 B]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [195 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [110 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [389 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [122 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [602 kB]
Get:22 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [125 kB]
Get:23 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [21.0 kB]
Get:24 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [4168 B]
Get:25 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Fetched 5057 kB in 25s (204 kB/s)
190 packages can be upgraded. Run 'apt list --upgradable' to see them.
harshika@siri:~$
```

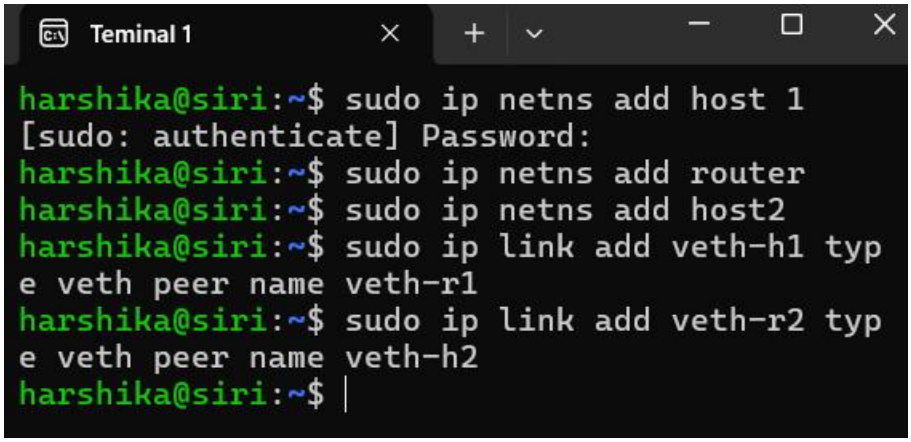
STAGE 2 — Build the network

Open again ubuntu and rename that tab as Terminal 1 and run all of these one by one:

sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2

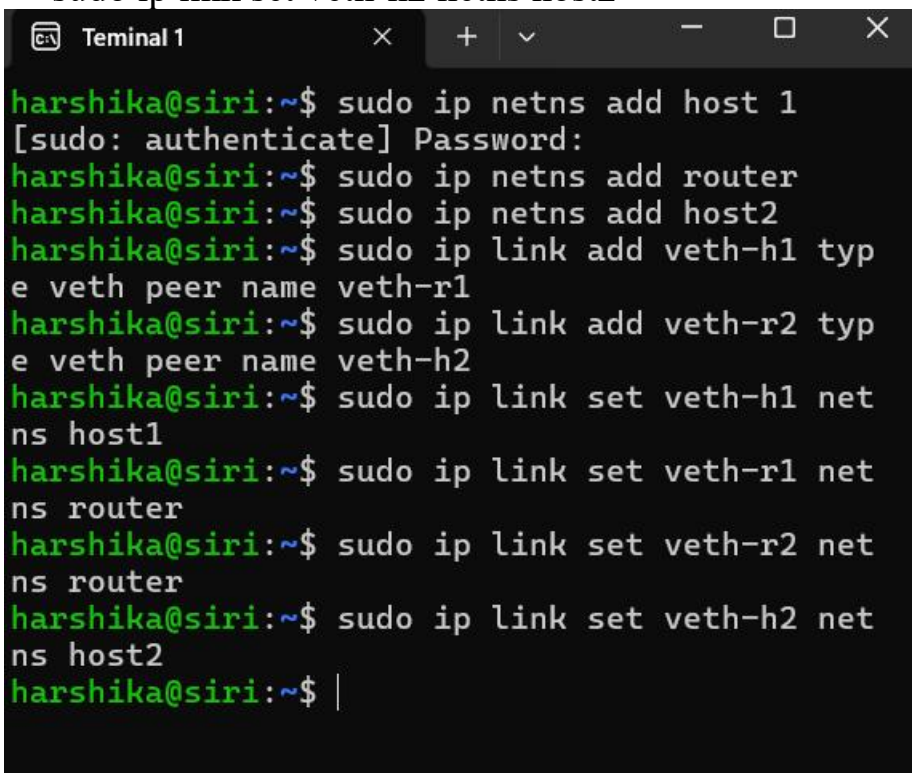


sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2



sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router

sudo ip link set veth-h2 netns host2

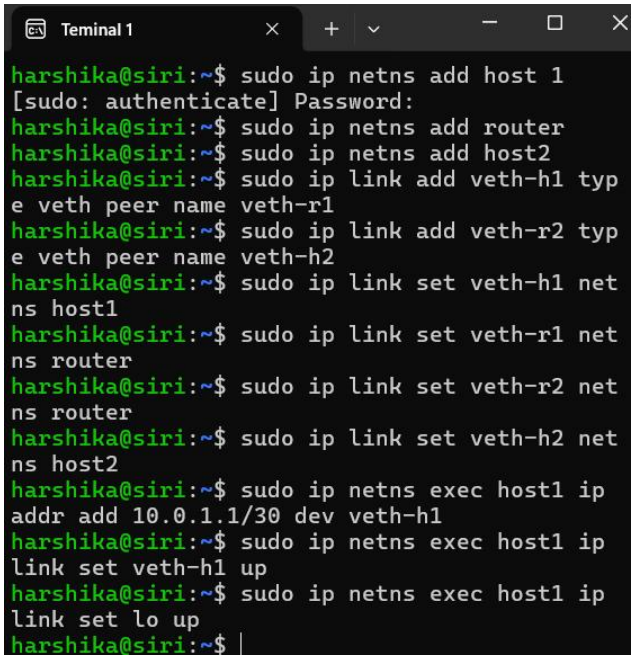


STAGE 3 — Configure IP addresses (Note: In terminal 1 Only) host1:

sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1

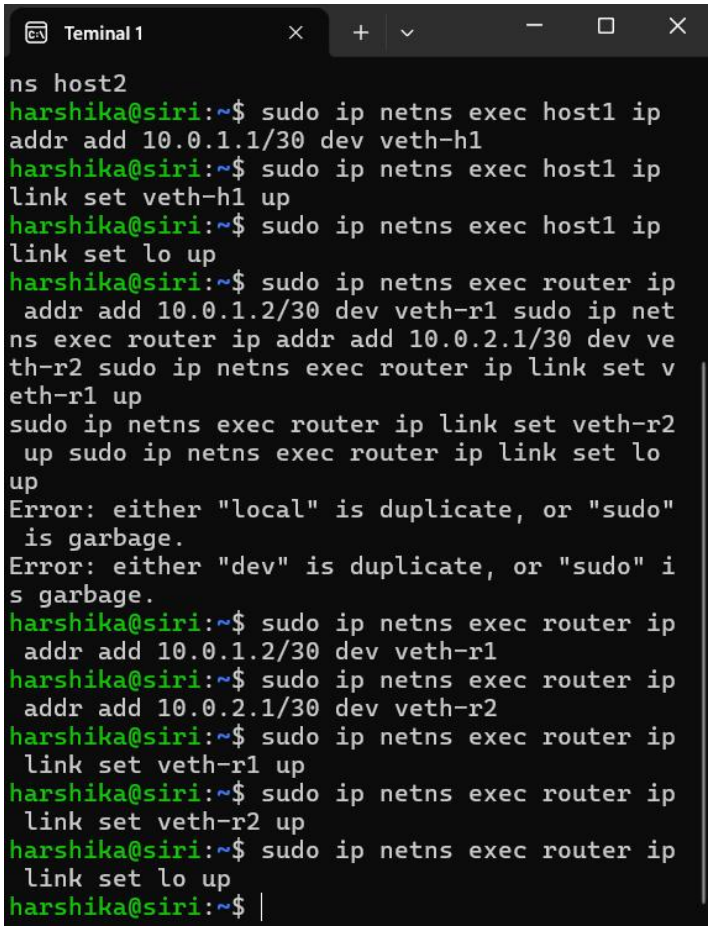
sudo ip netns exec host1 ip link set veth-h1 up

sudo ip netns exec host1 ip link set lo up



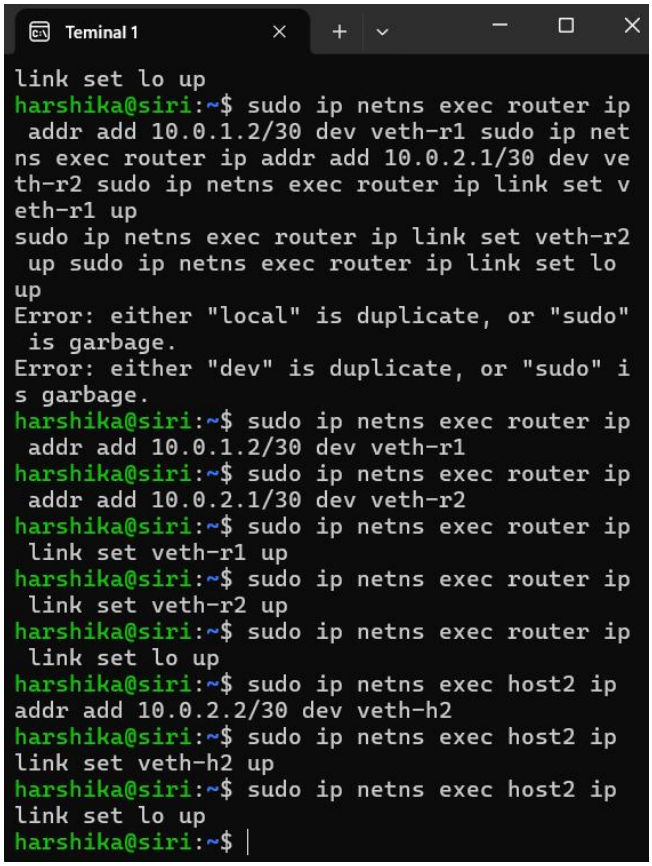
router:

sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up



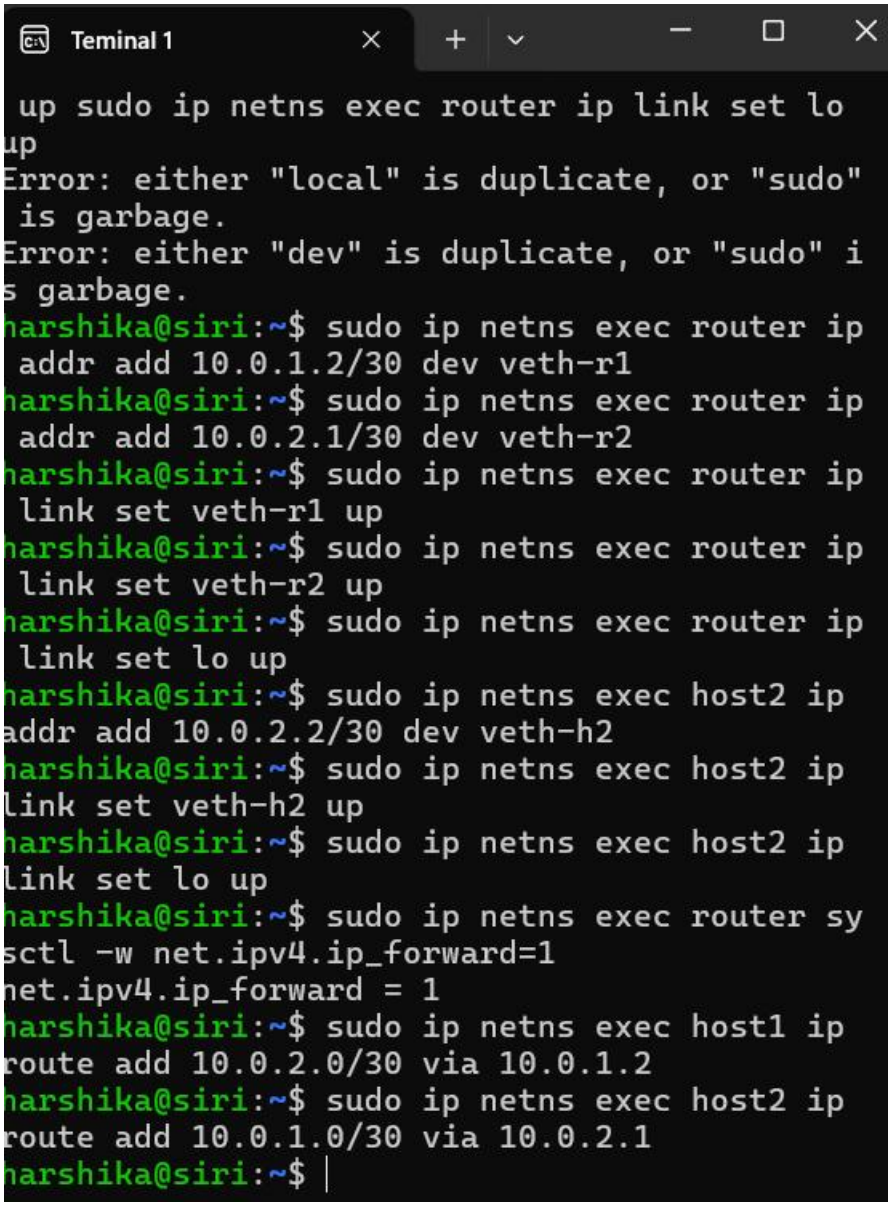
host2:

sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up


Teminal 1
link set lo up
harshika@siri:~$ sudo ip netns exec router ip
addr add 10.0.1.2/30 dev veth-r1 sudo ip net
ns exec router ip addr add 10.0.2.1/30 dev ve
th-r2 sudo ip netns exec router ip link set v
eth-r1 up
sudo ip netns exec router ip link set veth-r2
up sudo ip netns exec router ip link set lo
up
Error: either "local" is duplicate, or "sudo"
is garbage.
Error: either "dev" is duplicate, or "sudo" i
s garbage.
harshika@siri:~$ sudo ip netns exec router ip
addr add 10.0.1.2/30 dev veth-r1
harshika@siri:~$ sudo ip netns exec router ip
addr add 10.0.2.1/30 dev veth-r2
harshika@siri:~$ sudo ip netns exec router ip
link set veth-r1 up
harshika@siri:~$ sudo ip netns exec router ip
link set veth-r2 up
harshika@siri:~$ sudo ip netns exec router ip
link set lo up
harshika@siri:~$ sudo ip netns exec host2 ip
addr add 10.0.2.2/30 dev veth-h2
harshika@siri:~$ sudo ip netns exec host2 ip
link set veth-h2 up
harshika@siri:~$ sudo ip netns exec host2 ip
link set lo up
harshika@siri:~$

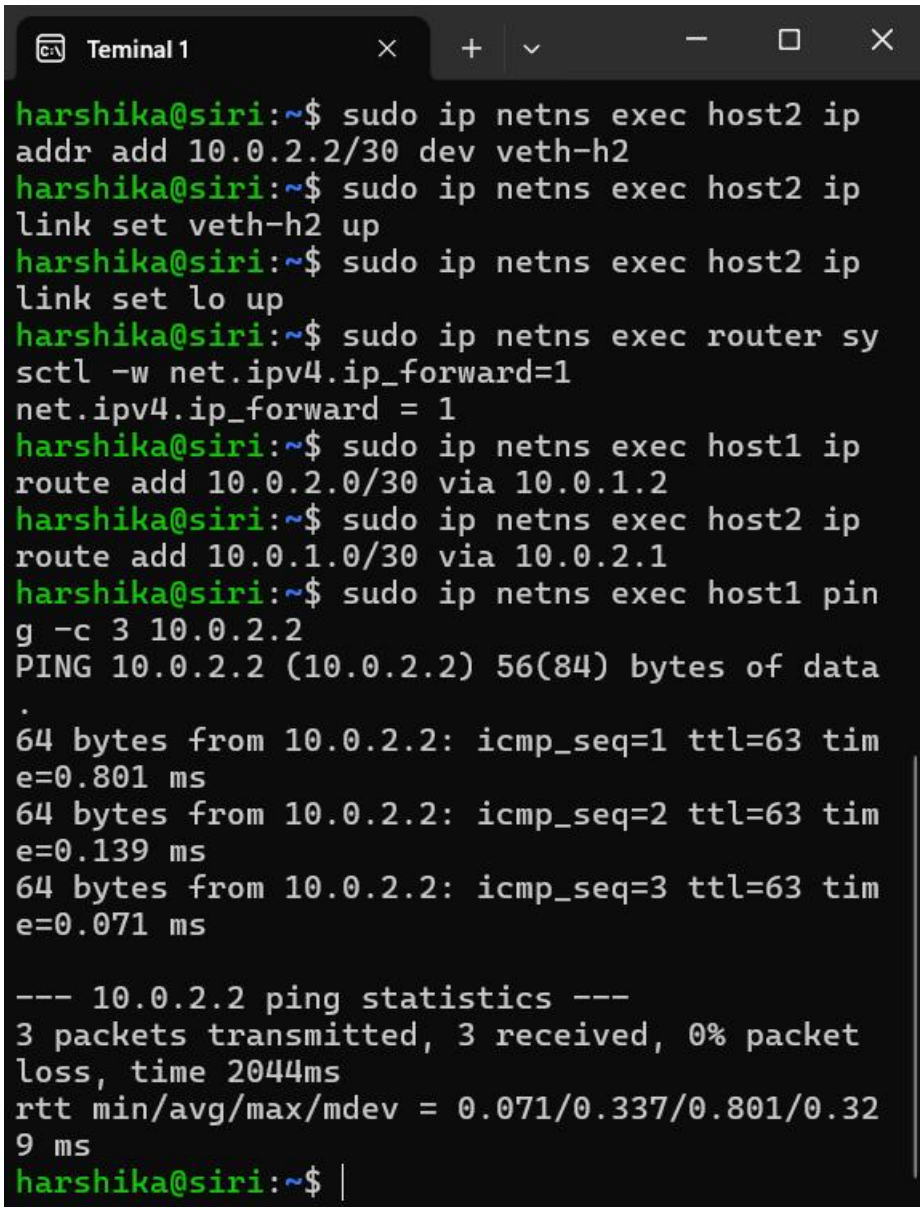
STAGE 4 — Enable routing

sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1



STAGE 5 — Test connectivity

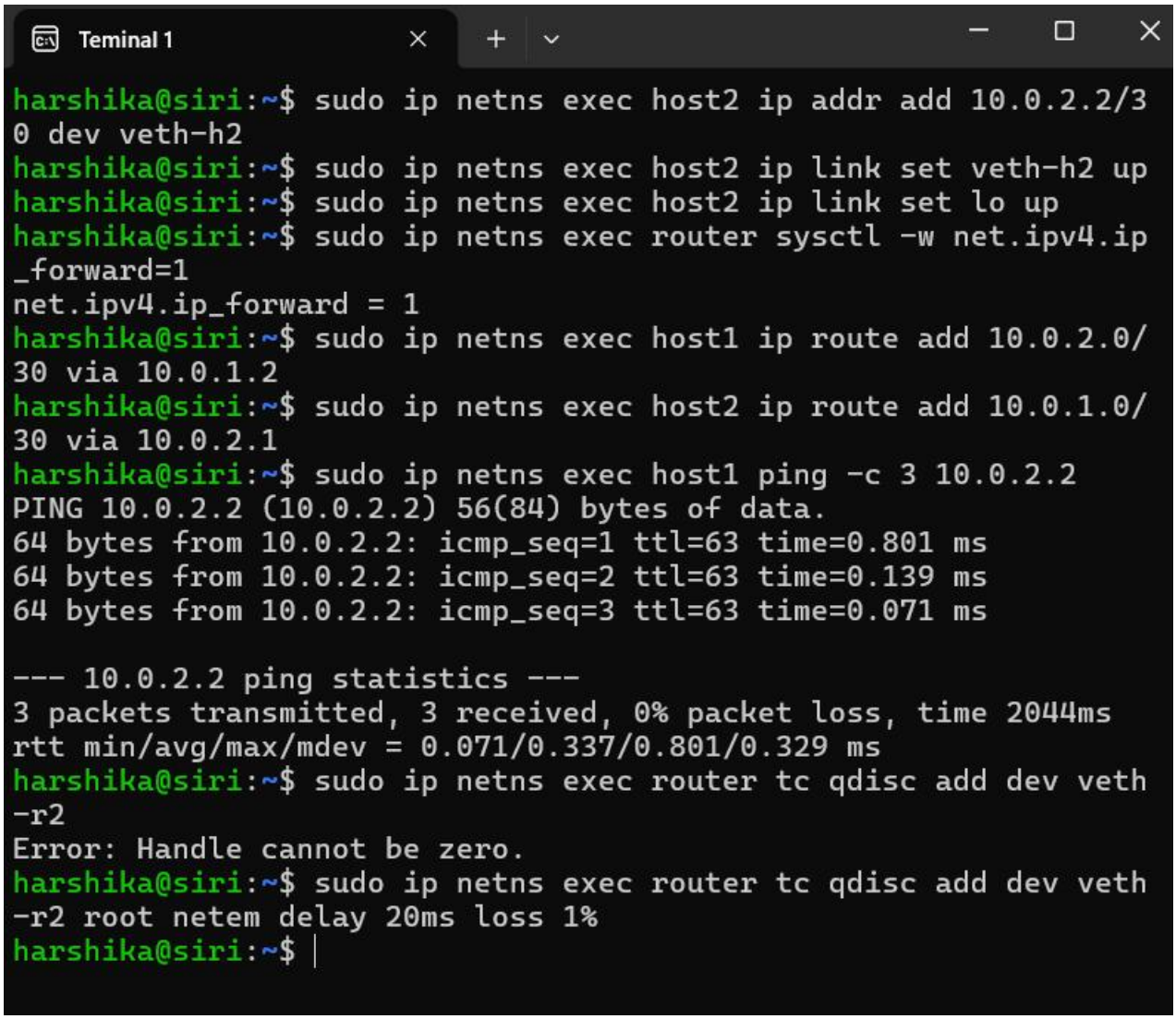
sudo ip netns exec host1 ping -c 3 10.0.2.2



You must see:

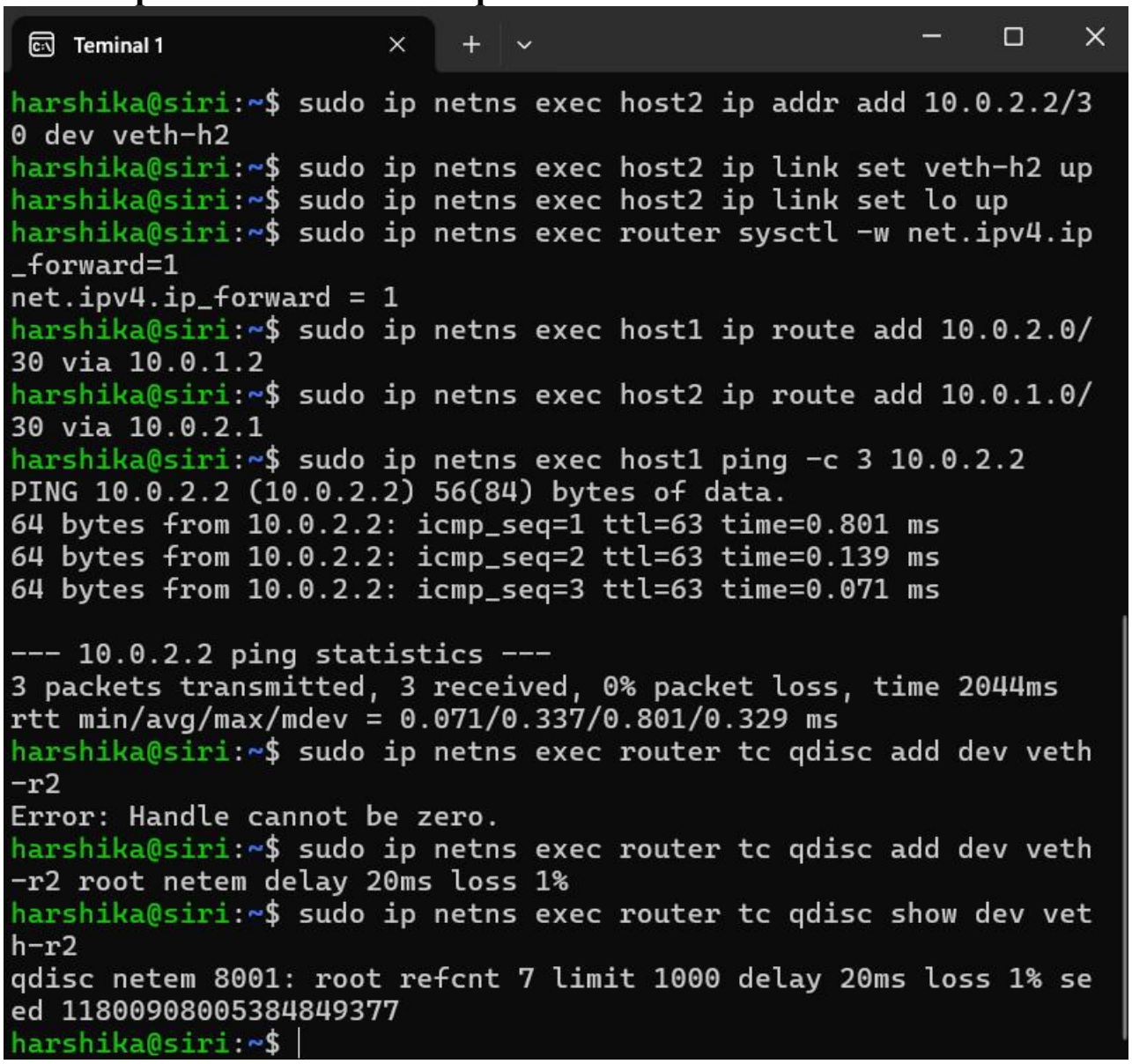
3 packets transmitted, 3 received, 0% packet loss

STAGE 6 — Add delay and loss



Verify:

sudo ip netns exec router tc qdisc show dev veth-r2

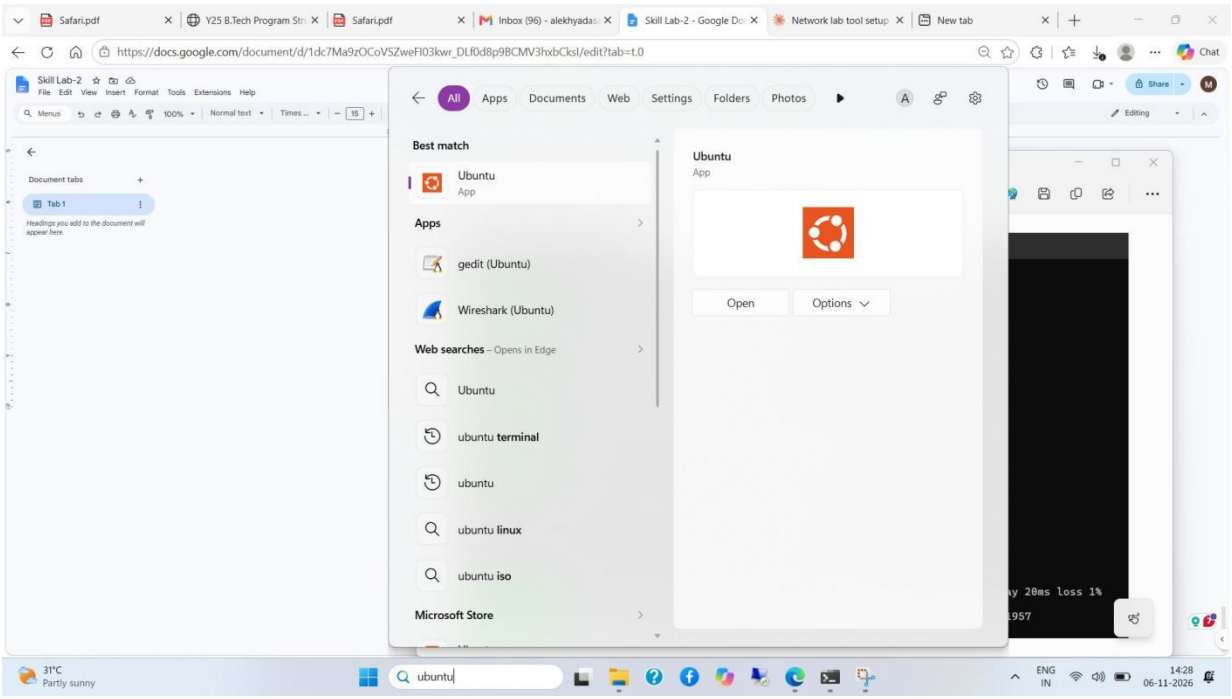


You must see netem delay 20ms loss 1%

STAGE 7 — Open 4 terminals

You need 4 terminals open now.

Terminal 1 — already open (used above)
Terminal 2 — open new (By going to Ubuntu option)

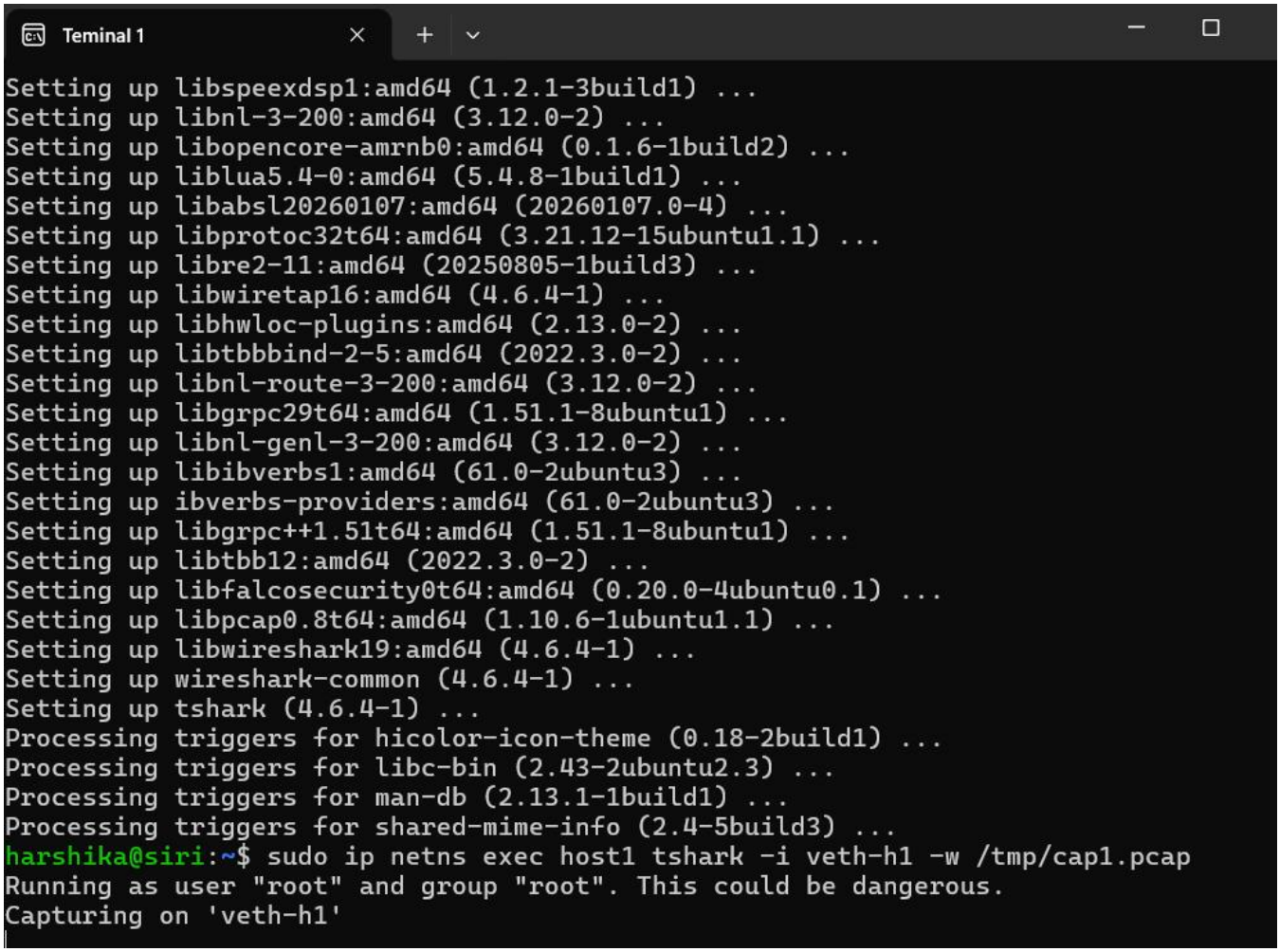


Terminal 3 — open new (By going to Ubuntu option)
Terminal 4 — open new (By going to Ubuntu option)

STAGE 8 — Start three captures simultaneously

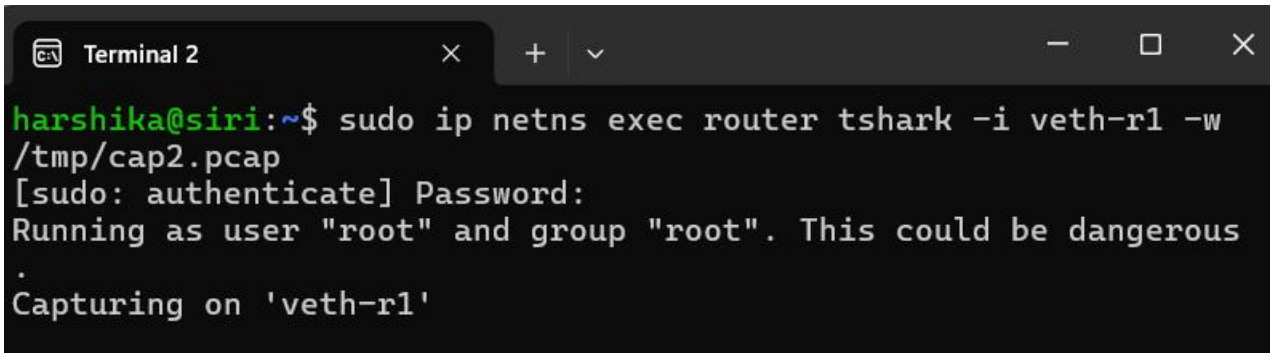
Terminal 1 — capture at host1

sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap



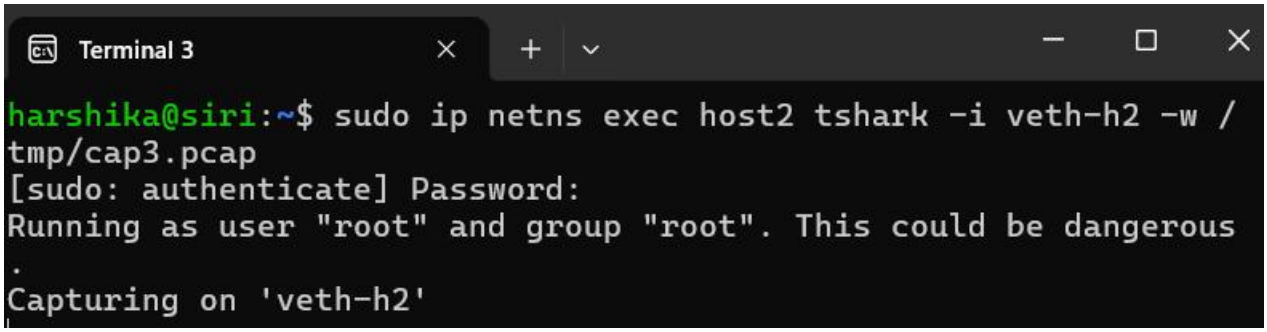
Terminal 2 — capture at router

sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap



Terminal 3 — capture at host2

sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap



All three show:

Capturing on 'veth-h1'

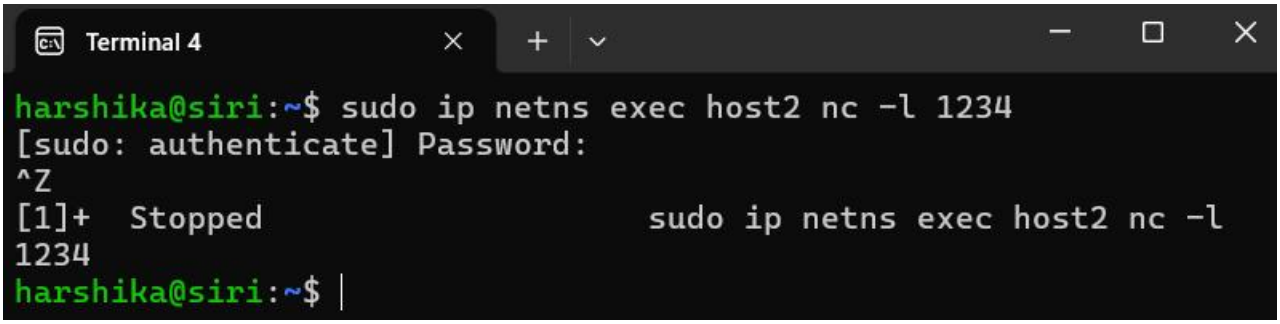
Leave all three running.

STAGE 9 — Send real message from host1 to host2

Go to Terminal 4.

First start the receiver on host2:

sudo ip netns exec host2 nc -l 1234



Open a **5th terminal.**

Connect from host1:

sudo ip netns exec host1 nc 10.0.2.2 1234

Now type this message and press Enter:

Hello from host1!

Type more messages:

This is Alekhya

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Three Point Capture Experiment

Message traveling through router

STAGE 10 — What you see in real time

Terminal 4 (host2) shows your message arriving:

Hello from host1!

This is Alekhya

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Three Point Capture Experiment

Message travelling through router

Terminal 1, 2, 3 (tshark) show packet counts increasing:

Capturing on 'veth-h1'

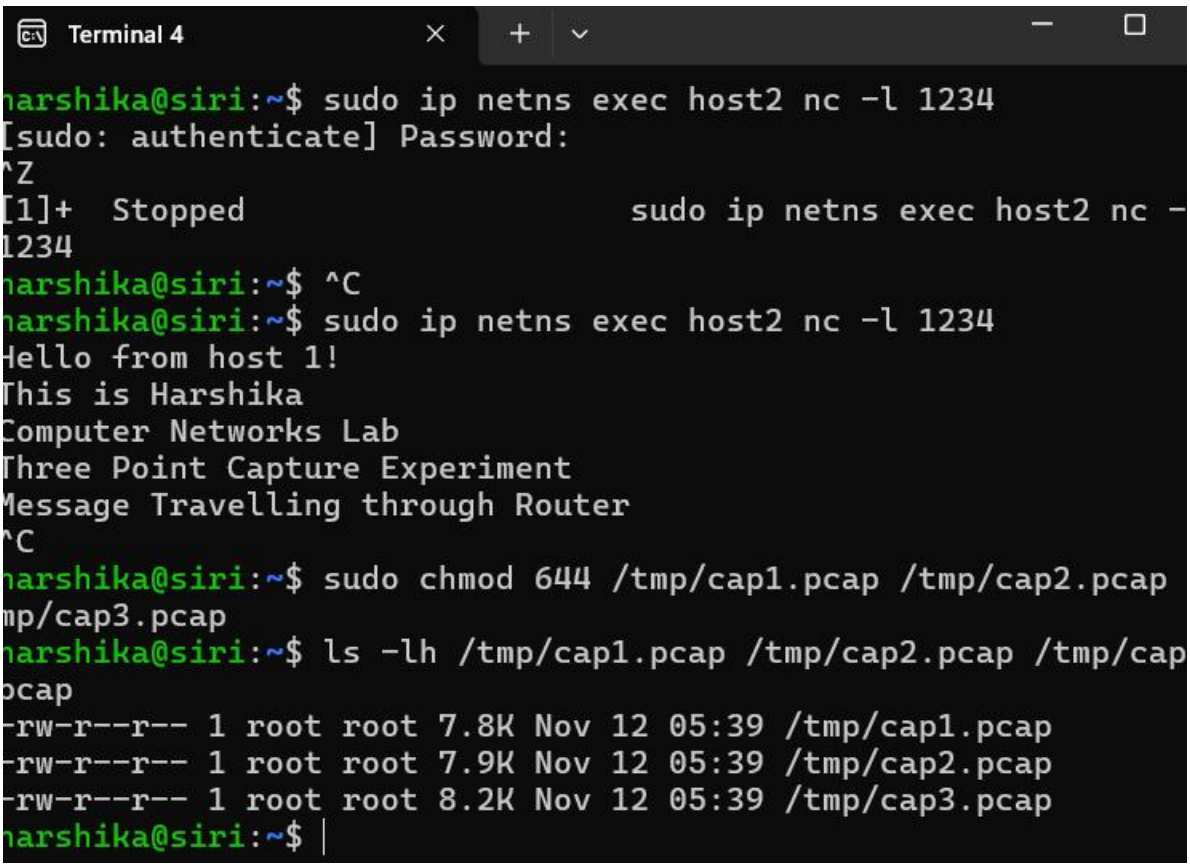
10 packets

24 packets

31 packets

STAGE 12 — Fix permissions

sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap

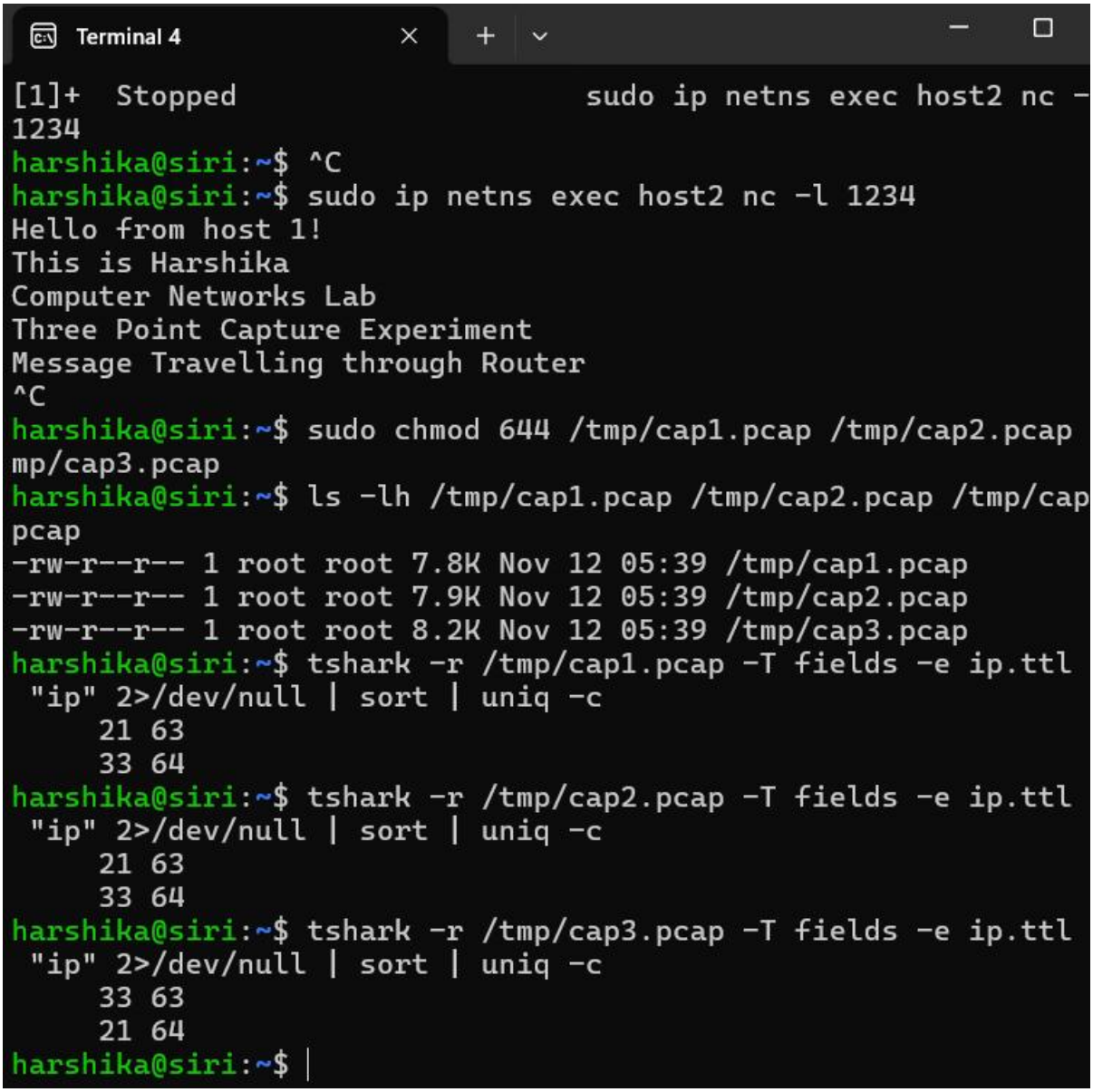


STAGE 13 — Check TTL

tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c

tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c

tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c



Expected:

cap1 → 64

cap2 → 64

cap3 → 63

STAGE 14 — Read your actual message from the capture

tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e data.text 2>/dev/null

You will see your actual message text extracted from the packet:

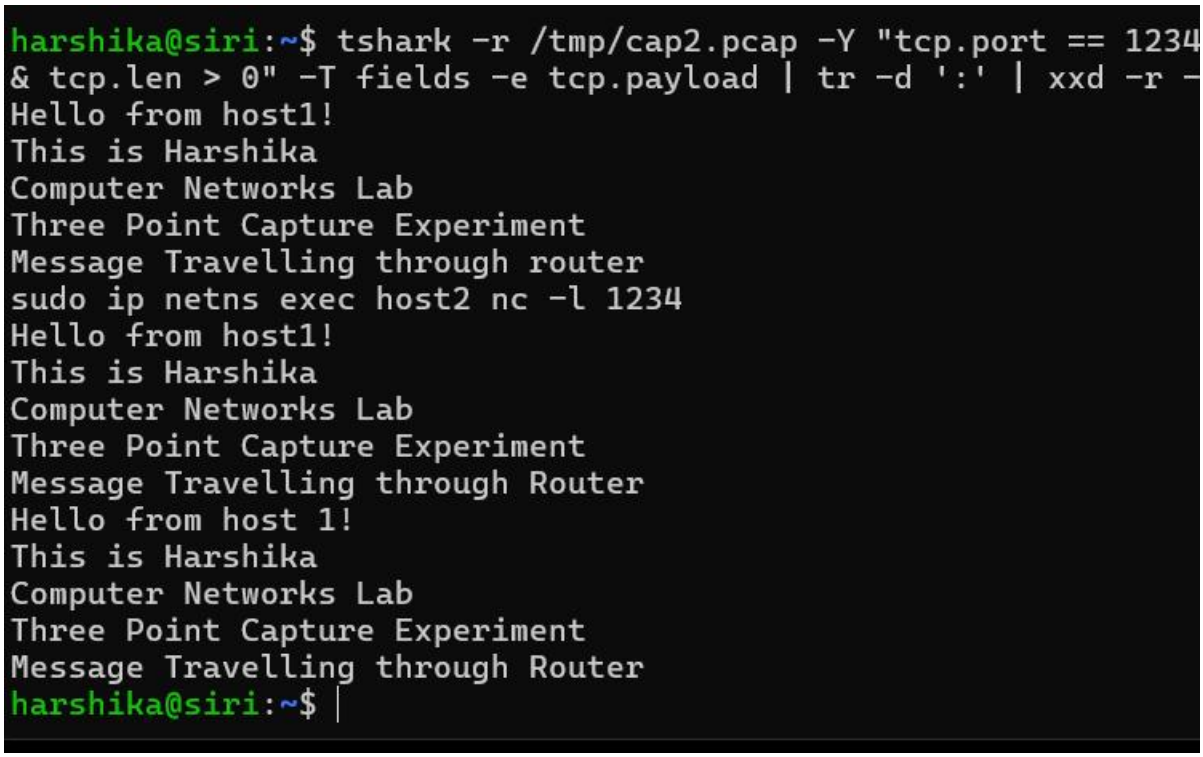
Hello from host1!

This is Alekhya

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Three Point Capture Experiment

Message travelling through router



Connect from host1:

TERMINAL 1 — Setup (run all commands here)

Install tools:

sudo apt install -y tshark iproute2 netcat-openbsd wireshark

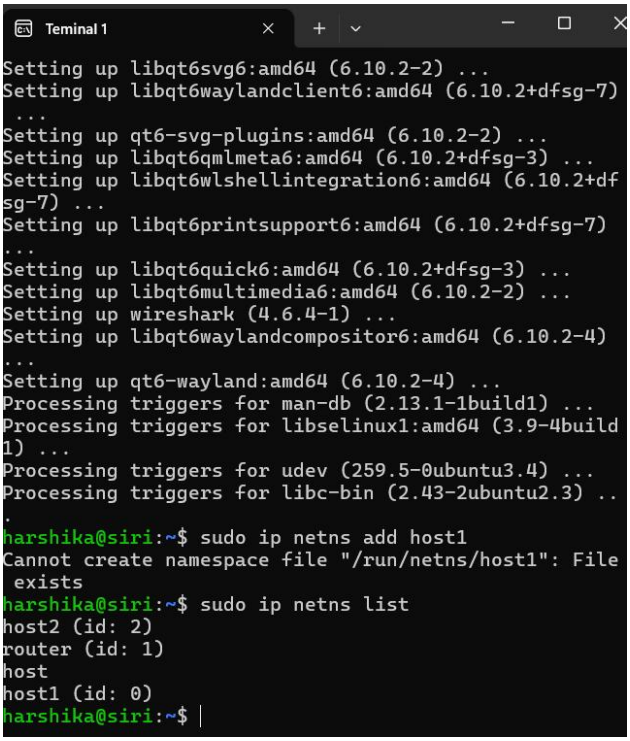
When asked → choose **Yes**

Create 3 nodes:

sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2

Verify:

sudo ip netns list



You must see: host1 router host2

Create cables:

sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2

